GeoReasoner: Geo-localization with Reasoning in Street Views using a Large Vision-Language Model

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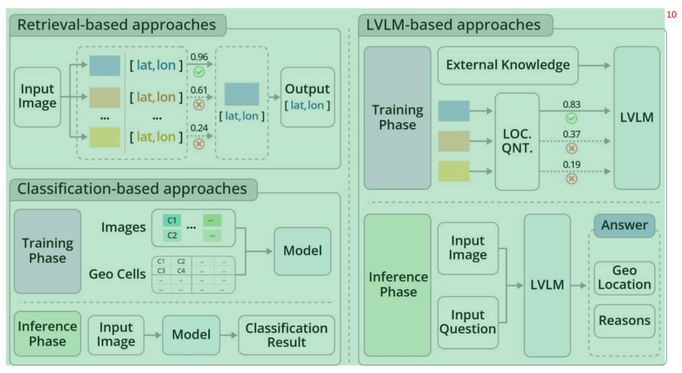
Introduction

This work tackles the problem of geo-localization with **a new paradigm** using a large vision-language model (LVLM) augmented with human inference knowledge.

Challenge: The scarcity of data for training the LVLM - existing street-view datasets often contain numerous low-quality images lacking visual clues, and lack any reasoning inference.

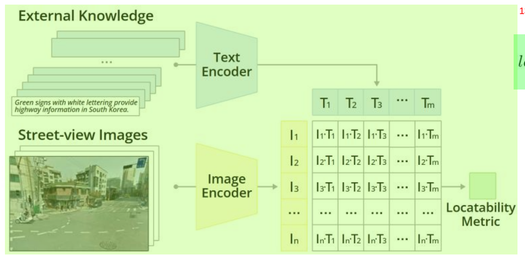
Dataset: We devise a CLIP-based network to quantify the **locatability** of street-view images, resulting in the creation of a new dataset comprising highly locatable street views. Additionally, we integrate external knowledge obtained from real geo-localization games, tapping into valuable human inference capabilities.

Model: **GeoReasoner**, which undergoes fine-tuning through dedicated reasoning and location-tuning stages, outperforms counterpart LVLMs by more than 25% at country-level and 38% at city-level geo-localization tasks, and surpasses StreetCLIP in performance while requiring fewer training resources.



Data Curation

Street-View Images: A total of over 130k street-view images with geo-tags collected from Google Street View
Textual Clues: A total of over 3K textual clues that encapsulate rich geo-localization information from *GeoGuessr* and *Tuxun* communities

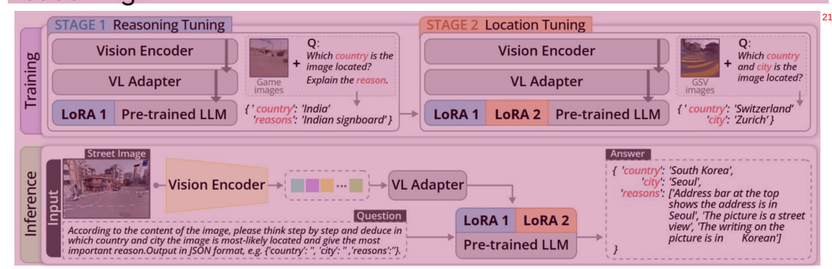


$$locatability(\mathbf{I}_{seg}, \mathbf{w}_{loc}) = \sum_{k=1}^{n} \mathbf{I}_{seg}(k) \cdot \mathbf{w}_{loc}^{k}$$

The locatability quantization network devises a CLIP-based visual-text pairing approach to predict the locatability metric.

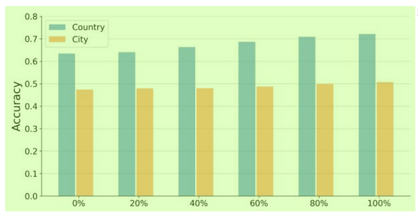
Geo-localization with Reasoning

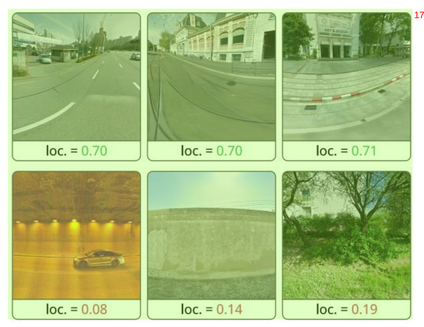
The architecture of **GeoReasoner** consists of three modules: Vision Encoder, VL Adapter and Pre-trained LLM. The model undergoes a two-fold supervised fine-tuning process: reasoning tuning and location tuning, to enable geo-localization with reasoning.



Experiments

Locatability Quantization Method





GeoReasoner

Comparison of Precision, Recall and F1 scores in country-level and city-level geo-localization. * represents the model trained on high-locatability GSV images.

Model	Country Accuracy↑	Country Recall↑	Country F1↑	City Accuracy↑	City Recall↑	City F1↑
StreetCLIP (Haas et al., 2023)	0.7943	1.00	0.8854	0.7457	1.00	0.8543
LLaVA (Liu et al., 2024)	0.4029	1.00	0.5744	0.2400	1.00	0.3871
Qwen-VL (Qwen-7B) (Bai et al., 2023a)	0.5829	0.95	0.7225	0.3743	0.89	0.5270
GPT-4V (Achiam et al., 2023)	0.8917	0.34	0.4923	0.5083	0.31	0.3851
ViT* (Dosovitskiy et al., 2021)	0.7100	1.00	0.8304	0.6762	1.00	0.8068
GeoReasoner*	0.8237	1.00	**0.9033**	0.7521	1.00	**0.8585**

Results of the ablation experiments

Model	Training: Reasoning	Training: Location	Performance: Country Accuracy↑	Country Recall↑	Country F1↑	City Accuracy↑	City Recall↑	City F1↑
Qwen-VL (Qwen-7B)	-	-	0.5829	0.95	0.7225	0.3743	0.89	0.5270
GeoReasoner w/o location tuning	✓	×	0.6971	1.00	0.8215	0.4114	0.99	0.5813
GeoReasoner w/o reasoning tuning	×	✓	0.7803	1.00	0.8766	0.7029	1.00	0.8255
GeoReasoner	✓	✓	**0.8237**	1.00	**0.9033**	**0.7521**	1.00	**0.8584**

Contributions

- We present a new paradigm that leverages an LVLM and external knowledge of human inference for geolocalization with reasoning from street-view images.
- We introduce the concept of locatability and devise a CLIP-based network to quantify the degree of locatability in street-view images.
- We propose GeoReasoner, an LVLM that outperforms existing geo-localization models and provides detailed reasoning for the inferred results.